POSITION PAPER · PREPRINT · NOT PEER REVIEWED

# The construction of OM: voice, breath and the scaffolding of meditation

Eugene Jhong · The OM Project, independent · 6 November 2026 · draft 2

Written by AI (Claude Opus 5.5, Anthropic) under the author's direction, and refereed by a second model (Claude Fable 5.1). See Authorship and AI use.

DOI: 10.5281/zenodo.24021098 · Correspondence and objections: github.com/ejhong/om/issues · PDF · The OM Project

## Abstract

OM is usually heard as one long tone. We propose that it is a constructed sound, and that the breath beneath it is the scaffolding on which meditation and breathwork are built.

The sound is a fast sequence of short notes that fuse into one. The notes come in four styles of vocalising (talking, rapping, yodelling and singing), with and without vibrato. Each note is made in one of the several physically different ways the voice has of producing the same pitch. The notes are arranged in eight sections, some built directly from notes and the rest permutations of those. Together they trace two lines: a drone on a home note, and a curve that rises from below it to just above, holds and returns.

There are four types of OM, set by register and voice quality. Each may serve a different area, with a feeling of its own: adaptation (bliss), bodily function (nonduality), the mind (gratitude) and the soul (loving-kindness).

Beneath the sound lies the breath. Athletics, vocalisation, meditation and breathwork all rest on a complex, subtle construction of the breath: thousands of tiny in- and out-breaths, with pressures in the body sensed by proprioception. Meditation, on this account, inverts the breath of OM-structured singing and the breath of athletic movement, and mixes the two; the meditative OM is silent. The type of OM sets the area, and the inverted athletic breath tunes the effect to a function within it.

If this is right, every meditation and breathwork practice builds on one scaffolding. People who are both skilled singers and skilled athletes may reach deeper meditative states, provided they also build OM's structure. We state each claim and what it rests on, set out the rival accounts, and give the measurements that would refute it.

> **A caution.** This paper describes; it does not prescribe. Nothing in it is a practice or a protocol. Breathwork, cold exposure and intensive meditation carry real risks (Section 12). Anyone in difficulty can find support from Cheetah House (cheetahhouse.org).

## 1. Introduction

### 1.1 OM

OM is chanted across the Hindu, Buddhist and Jain traditions, alone or to open and close a practice. The Mandukya Upanishad begins by naming everything as OM: what was, what is, what will be, and whatever lies beyond. It then divides the syllable into three sounds, A, U and M, with a fourth beyond sound (Mandukya Upanishad, verses 1 and 8–12, in Olivelle 1998).

In practice OM is chanted as one long tone on a single breath, and it has been studied as such. In a pilot brain-imaging study of twelve volunteers, chanting OM aloud lowered activity in limbic regions compared with rest, and saying "ssss" did not (Kalyani et al. 2011). The rosary and a yoga mantra, each recited six times a minute, both strengthened the natural rhythms of the heart and circulation, and the baroreflex (Bernardi et al. 2001).

### 1.2 A constructed sound

We propose that the long tone is a construction. Heard as a whole, it is one sound. But it is built from many short notes made in different ways, arranged in a definite structure, and carried by a breath that is itself constructed. We also propose that the same breath, inverted, is what meditation and breathwork are made of.

### 1.3 Where the hypothesis came from

The hypothesis comes from the author's own practice. The OM Project (ejhong.github.io/om) renders one version of the construction as a score and a player with a synthesised voice. Four recordings of the author's OMs, one of each type, are the only data so far (Section 3.6).

A companion paper proposes what the body's knots are (Jhong 2026). On the present account, releasing them is one of the things this breath does (Section 4).

---

## 2. The hypothesis

> *OM is a constructed sound, and its breath is the scaffolding of meditation and breathwork. The sound is a fast sequence of short notes. They are made in four styles, with and without vibrato, each by one of the voice's several ways of producing the same pitch, and they are arranged in eight sections that trace a drone and a curve. Its breath, like the breath of all athletics, vocalisation, meditation and breathwork, is built from thousands of tiny in- and out-breaths and pressures in the body. Meditation inverts the breath of OM-structured singing and of athletic movement and mixes them. The type of OM sets the area of its effect, and the athletic breath sets the function within it.*

In parts, each of which can be checked on its own:

1. **Sequence.** OM is a fast sequence of short notes that fuse into one sound. *(hypothesis)*
2. **Ingredients.** Its notes come in four styles of vocalising (talking, rapping, yodelling, singing), with and without vibrato. The voice has several physically different ways of producing the same pitch, and which one a note uses matters. *(the voice's abilities: established; their use in OM: hypothesis)*
3. **Structure.** The notes are arranged in eight sections. Some are built directly from notes; the rest are permutations of those. *(hypothesis)*
4. **Two lines.** The sequence traces a drone on a home note, and a curve that rises from below it to just above, holds and returns. *(hypothesis; a curve of this kind is seen in one person's recordings)*
5. **Four types.** Register (modal or falsetto) and voice quality (normal or operatic) give four types of OM. Each serves an area, with a feeling of its own. *(hypothesis)*
6. **Breath construction.** Athletics, vocalisation, meditation and breathwork rest on a construction of the breath: thousands of tiny in- and out-breaths, with pressures in the body sensed by proprioception.

*(the breath's coupling to action: established; its fine construction: hypothesis)*

7. **Inversion.** In singing and athletics the act rides the exhale. Meditation inverts the breath of OM-structured singing and of athletic movement and mixes them. The type of OM sets the area; the inverted athletic breath tunes the function within it. *(hypothesis)*
8. **Scaffolding.** Every meditation and breathwork practice builds on this construction, whether or not it uses the chant. *(hypothesis)*
9. **Skill and structure.** Singing and athletic skill give better material for building OM, but its structure has to be practised too. Skilled singer-athletes who build it may reach deeper meditative states. *(hypothesis)*

---

## 3. How OM is built

This section states the construction as the hypothesis has it; Section 8 separates what is established from what is not.

### 3.1 Four styles of vocalising

Talking, rapping, yodelling and singing are four ways of using the same instrument. Each coordinates the breath, the larynx and the mouth in its own way:

- **talking**, with the varying pressure and pitch of speech;
- **rapping**, with fast, rhythmic bursts;
- **yodelling**, with rapid switches between the voice's two main registers;
- **singing**, with a sustained, supported tone.

OM uses all four, and the hypothesis is that each contributes something the others do not.

### 3.2 Vibrato and straight tone

Vibrato is a regular wave in pitch. In classical singing it runs at about five to six cycles a second: over three years of training, students whose vibrato was slower than 5.2 cycles a second sped up, and those faster than 5.8 slowed down (Mürbe et al. 2007). A straight tone holds its pitch without it. OM combines the two.

### 3.3 One pitch, several ways

The voice can produce the same pitch in physically different ways:

- **In either of two modes of vibration.** The larynx vibrates in distinct mechanisms. The two main ones correspond to the modal voice of speech and the lighter falsetto, and their ranges overlap, so over a band of pitches a note can be sung in either (Roubeau et al. 2009).
- **With either of two qualities.** A note can be sung with the bright, forward quality of speech, or with the darker, ringing quality of operatic singing. The operatic quality gathers energy into a peak near 3 kHz, the "singer's formant", made by a clustering of the voice's higher resonances (Sundberg 2001).

The results sound roughly alike but are made differently. The hypothesis is that the choice is not neutral: which way a note is made matters, and OM chooses. This is one ingredient among several, on the same footing as the four styles.

### 3.4 A fast sequence

OM is not a single held setting of the voice. It is a fast sequence of short notes, each made in its own way and too short to hear one by one, so they fuse into one sound, as frames fuse into motion. In the project's rendering, each note lasts a few hundredths of a second. We call this fusion blending: the listener hears one OM, and the singer makes many notes.

### 3.5 Eight sections

The sequence unfolds in eight sections. Some are built directly from notes. The others are permutations of those sections, recombining their notes in new orders, so that later material is earlier material rearranged. Whether the simpler sections come first or last is open, and it may differ between the four types of OM.

### 3.6 Two lines

Heard over its few seconds, the sequence traces two lines (Figure 1a):

- **the drone**, which holds a home note;
- **the curve**, which starts below the home note, rises to just above it, holds and returns.

Pitch tracking of four recordings of the author's OMs, one of each type, follows a curve of this kind (Figure 1b). Taking the note each recording ends on as its home note, the pitch starts two to three semitones below it and rises within about a second to the home note or a little above it: about a semitone above in smad, less in the others. It holds there through the middle.

Part of this may simply be how any voice enters and leaves a held note. One recording of each type is an example, not a measurement. The tracking was also smoothed, and a smoothed track can only show the two lines blended. Whether the drone is there is a question for finer tracking (Prediction 3). The hypothesis is that both lines are part of the construction.

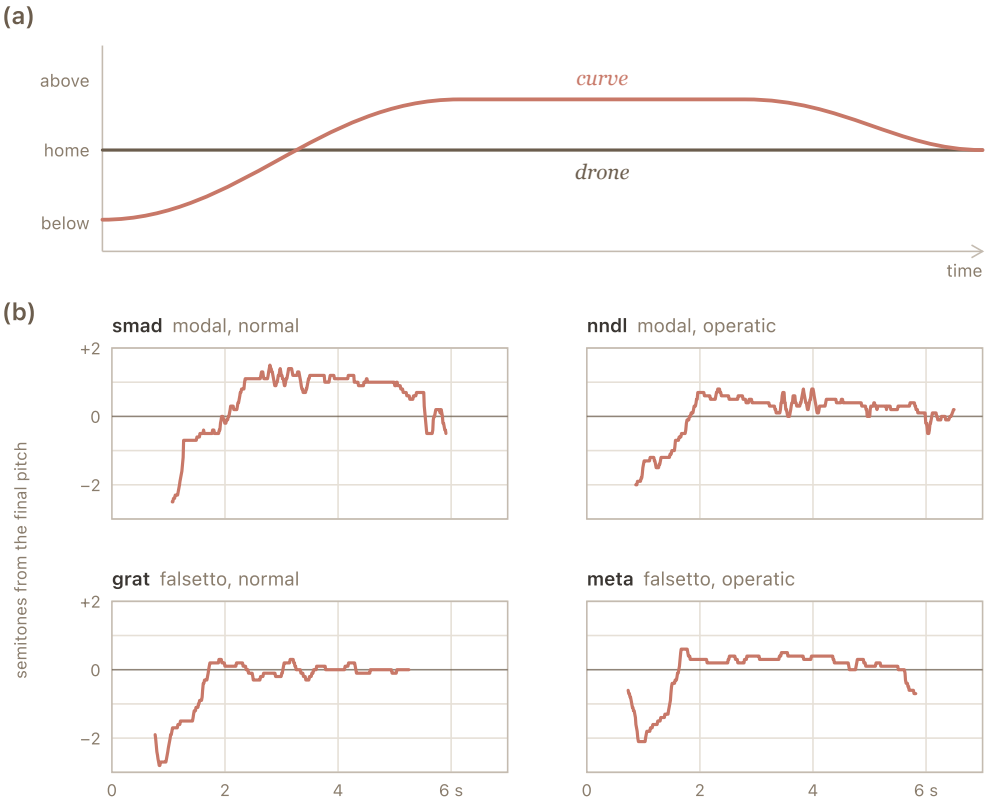



**Figure 1.** (a) The hypothesis, drawn plainly: a drone on the home note, and a curve that rises from below it, holds just above and returns. (b) Pitch of four recorded OMs by one practitioner, one of each type, in semitones from each recording's final pitch (pYIN pitch tracking, smoothed with a running median over about a fifth of a second; in two recordings, a burst of under a tenth of a second before the note begins, which the tracker placed an octave low, is left out). An example, not a measurement.

## 4. Four types of OM

Two choices in how the voice is used give four types of OM: the register (modal or falsetto) and the voice quality (normal or operatic). The hypothesis is that each type serves a different area, and that each brings a feeling of its own (Table 1).

The names are rough abbreviations, after the feelings: smad of *samadhi*, the absorption whose mark is bliss; nndl of *nondual*; grat of *gratitude*; and meta of *metta*, the Pali word for loving-kindness. The areas and feelings are rough sketches of possible categories, not a map.

**Table 1.** The four types of OM.

| Type | Register, quality | Area | Examples | Feeling |
|---|---|---|---|---|
| smad | modal, normal | adaptation | cold and heat; releasing knots; relaxation, as in a body scan; resisting pain and motion sickness | bliss |
| nndl | modal, operatic | bodily function | bowels, bladder, clearing the sinuses, sex | nonduality |
| grat | falsetto, normal | the mind | reaction time in sport; games such as chess; creativity; reading and writing (the different modes a mind falls into) | gratitude |
| meta | falsetto, operatic | the soul | healing, love | loving-kindness |

The area is where the effect is aimed; the feeling is how the type is felt from inside. Within each area, the breath tunes the function further (Section 5.3).

---

## 5. The breath

### 5.1 Breath construction

On this account, athletics, vocalisation, meditation and breathwork all rest on a complex and subtle construction of the breath. It has two parts:

- **A fine structure of thousands of tiny in- and out-breaths.** They can ride on what feels like a single ordinary inhale or exhale, yet the one making them can clearly feel them.
- **A set of pressures in the body,** held and moved, and sensed by proprioception.

The breath as it is usually described (a rate, a depth, a ratio of in to out) is only the coarse outline of this construction.

Some of this is established:

- **Breathing locks to movement.** Running mammals synchronise breath and stride. Human runners use several locked ratios of strides to breaths, most often two to one (Bramble and Carrier 1983).
- **The diaphragm breathes and braces at once.** During repeated movement of an arm, the diaphragm and the deepest abdominal muscle held the pressure in the abdomen up. Meanwhile their activity rose and fell with both the breath and the movement (Hodges and Gandevia 2000).
- **Effort rides the exhale.** Collegiate tennis players served and hit forehands faster, and pushed harder in held strokes, when they grunted (O'Connell et al. 2014).
- **Singers build their breath.** In classical singers, the abdomen holds a posture that sets the diaphragm and rib cage, and the rib cage regulates the pressure that drives the voice (Watson and Hixon 1985). Professional opera singers repeated the same pattern of lung volume closely each time they sang a phrase, though the shares of the rib cage and the abdomen in it differed from singer to singer (Thomasson and Sundberg 1999).
- **Contemplative traditions pair the breath with pressure in the body.** Hatha yoga's bandhas, or locks, contract the throat, the abdomen and the root of the trunk, typically while the breath is held (Mallinson and Singleton 2017).
- **The vase breath of tummo.** In Tibetan tummo's "vase" breath, as Kozhevnikov and colleagues describe it, the breath is held after an in-breath while the abdominal and pelvic muscles contract, pushing the lower belly out like a vase. In expert practitioners, the forceful form of the practice raised armpit temperature from normal into a slight to moderate fever (up to 38.3 °C). The breath alone, without the practice's visualisation, raised it less, within the normal range (Kozhevnikov et al. 2013).

What is new is the fine structure. To our knowledge, no one has measured tiny reversals of flow riding on the breath during any of these activities.

### 5.2 The felt phases

In singing and athletics, the felt breath has a familiar shape: one breathes in to get ready, and the act (the voice, or the burst of effort) rides the exhale.

In meditation it can be hard to tell which phase is which, and the effect can be made on either, because it rides on the tiny breaths rather than the large one. The author finds it comes most easily the other way round from performance: the construction is made on the exhale and activated on the inhale.

Published work fits a role for the inhale:

- Pain from brief shocks was rated lower near the end of the in-breath than of the out-breath, though the effect was small and a spinal pain reflex went the other way (Arsenault et al. 2013).
- In recordings from inside the brain in people with epilepsy, activity in the amygdala and hippocampus followed nasal breathing, strongest during the in-breath; in behavioural experiments, the breath's phase affected how well fear was recognised and memories recalled (Zelano et al. 2016).
- People spontaneously breathe in as they begin a mental task, and breathing in through the nose improved accuracy on a visuospatial task (Perl et al. 2019).

### 5.3 The inversion

Meditation, on this account, takes the breath of two trained skills, singing in OM's structure and athletic movement. It inverts each, turning each tiny breath the other way, and mixes them. Seen from the inside, this may be why meditation's felt phases tend to run the other way round from performance's (Section 5.2).

The meditative OM is silent: the construction is carried by the breath alone.

The two inverted breaths do different work:

- **The OM's breath sets the area,** by its type: adaptation, bodily function, the mind or the soul.
- **The athletic breath tunes the effect** to a particular function within that area: within adaptation, for instance, cold rather than pain.

Different movements carry different breath constructions, so the repertoire of athletic breath would be what divides each area into its many functions.

---

## 6. The scaffolding

If meditation is the inverted, mixed breath of OM and athletics, then every meditation and breathwork practice builds on that construction, whether or not it uses the chant. Each would be a partial or approximate form of it, reached by its own route.

The claim predicts that existing practices sort into the four areas.

- **By feeling:**
  - absorption practices whose mark is bliss (samadhi; the jhanas of Buddhist meditation) would sit with smad;
  - nondual traditions, such as Dzogchen and Advaita Vedanta, with nndl;
  - practices of gratitude with grat;
  - loving-kindness (metta) meditation with meta.
- **By function:** practices of cold and heat (ice baths, sauna, tummo) and body scans would sit with smad. So would the release of knots described in the companion paper (Jhong 2026), whose trained breath would be smad's.

The sorting is speculative. Its use is that it can be tested (Prediction 9): practices in the same area should share part of their breath construction.

---

## 7. Skill and structure

The construction asks a great deal of the voice and the breath:

- **Singing skill supplies the materials.** It gives command of the four styles, of both registers and both qualities, of vibrato and straight tone, of the right way to make each note, and of the breath that carries them.
- **Athletic skill supplies the breath of movement.** It gives a repertoire of breath constructions tied to movement, and fine control of pressure in the body.

Better singing and better athleticism should therefore build a more perfect OM. But neither supplies the structure: the sequence, the sections and their permutations have to be practised for themselves.

Hence the claim: skilled singers who are also skilled athletes may reach deeper meditative states than others, provided they also build OM's proper structure. Skill without the structure, or the structure without the skill, should go less deep.

---

## 8. The case

Most of this paper is hypothesis, and it should be read as such.

**What is established is that its materials exist:**

- The voice can produce one pitch in more than one way: in two mechanisms whose ranges overlap (Roubeau et al. 2009), and with or without the operatic singer's formant (Sundberg 2001).
- Vibrato is a regular wave in pitch whose rate training shapes (Mürbe et al. 2007).
- Skilled activity couples the breath to action and controls it finely (Bramble and Carrier 1983; Hodges and Gandevia 2000; O'Connell et al. 2014; Watson and Hixon 1985; Thomasson and Sundberg 1999).
- The breath's phase and rate shape experience and physiology:
  - pain (Arsenault et al. 2013; Zautra et al. 2010; reviewed by Jafari et al. 2017);
  - cognition (Zelano et al. 2016; Perl et al. 2019);
  - the body's sympathetic drive (Oneda et al. 2010; Adler et al. 2019).

**What fits the hypothesis, without testing it:**

- The rosary and a yoga mantra, recited six times a minute, strengthened the natural rhythms of the heart and circulation, and the baroreflex (Bernardi et al. 2001).
- In a pilot study, chanting OM aloud lowered limbic activity compared with rest, where saying "ssss" did not (Kalyani et al. 2011).
- Tummo's forceful breath, with its visualisation, raised expert practitioners' armpit temperature into a slight to moderate fever (Kozhevnikov et al. 2013).
- Ten days' training in meditation, a breathing method and cold exposure let volunteers raise their adrenaline at will and damp their immune response to an injected bacterial toxin (Kox et al. 2014).

- In one study, the same deep, slow breathing raised pain thresholds when done with relaxation, but not when done with effortful attention (Busch et al. 2012). How a breath is done can matter beyond its rate.

**What nothing yet shows:** that OM is built as proposed, that the breath carries a fine structure, that meditation inverts it, or that the four types do different things.

---

## 9. The alternatives

Four simpler accounts could explain what OM chanting and meditation do without any construction. Each deserves its strongest form.

1. **Slow breathing.** Chanting paces the breath, and slow breathing alone lowers sympathetic activity (Oneda et al. 2010; Adler et al. 2019) and eases pain (Zautra et al. 2010). The rosary and a yoga mantra, recited at the same pace, had a similar effect on the rhythms of the heart and circulation (Bernardi et al. 2001). On this account, any slow chant would do. Against it: the same deep, slow breathing raised pain thresholds when done with relaxation but not with effortful attention (Busch et al. 2012), so how a breath is done can matter beyond its rate; that difference, though, is one of attention, which the fourth account also claims.
2. **Breath chemistry.** Breathwork could act through the blood gases set by the rate and depth of breathing, rather than through any fine structure. In one trained method, cycles of hyperventilation and breath-holding brought on respiratory alkalosis and low blood oxygen, and with them a surge of adrenaline that damped the immune response (Kox et al. 2014).
3. **Vibration.** Chanting OM aloud is felt as vibration, and that vibration has been proposed to stimulate the vagus nerve through its branches to the ear (Kalyani et al. 2011). On this account, the hum does the work, not the construction.
4. **Attention and expectation.** Any focused practice that is expected to help may help.

Each of these predicts that a plain hum, matched in breath rate, loudness, length and what participants expect of it, should do what a well-built OM does. The construction hypothesis predicts it should not (Prediction 7).

---

## 10. Predictions, and what would refute them

**Table 2.** What the hypothesis predicts, how to look, and the result that would refute it.

| Prediction | How to look | Refuted if |
|---|---|---|
| 1. **Micro-breaths.** Thousands of tiny in- and out-breaths ride on the breath during skilled singing, athletics, meditation and breathwork. | Airflow at the mouth and nose sampled fast, with bands at the chest and belly, in skilled practitioners. | No small reversals of flow appear beyond those the heartbeat produces. |
| 2. **Inversion.** The fine structure of meditation contains the singer's, inverted. | The same person singing OM, then meditating with the silent OM: compare the micro-breath patterns. | The singer's pattern, inverted, is not found in the meditative one. |
| 3. **Two lines.** A skilled OM switches rapidly between a drone and a curve. | Pitch tracking at a resolution of a few milliseconds; the four recordings are a start. | The pitch moves as one line, with only ordinary jitter and vibrato. |
| 4. **The four types are sung as described.** | Acoustics (spectrum, singer's formant) and electroglottography of the four types as sung. | The recordings do not show the register and quality that Table 1 gives each type. |
| 5. **Each type brings its own feeling.** | After each type, participants who don't know the hypothesis rate bliss, nonduality, gratitude and loving-kindness. | No type peaks on its own feeling. |
| 6. **Each type serves its area.** | Measures matched to each area: tolerance of cold and pain (smad); nasal airflow (nndl); reaction time and performance in games and creative tasks (grat); self-rated compassion (meta). | The types do not differ on their own area's measures. |
| 7. **The construction matters.** | A well-built OM against a plain hum and against an OM with its sections scrambled, matched in breath rate, loudness and length, with participants blind to which is which. | They do the same. |
| 8. **Skill and structure.** | Singers, athletes, both and neither, before and after learning OM's structure; depth of meditation by questionnaire and physiology. | Learning the structure adds nothing for the skilled, or skill adds nothing once the structure is learnt. |
| 9. **Scaffolding.** | Record the fine breath in practices sorted to the four areas: tummo, body scan, nondual practice, gratitude, metta. | The fine structure does not cluster by area. |
| 10. **The inhale.** Activation tends to land on the inhale. | Time the onset of effects, felt and measured, against the breath. | Effects land as often, or more often, on the exhale. |

## 11. Limits

- The hypothesis comes from one practitioner's practice, and the reports behind it are the author's own.
- The fine structure of the breath is described from the inside and has not been recorded. Inside views of breathing can mislead: classical singers' accounts of how they breathed matched their measured breathing poorly (Watson and Hixon 1985).
- The four recordings are one take of each type by one person, and their pitch tracks are smoothed.
- The project's rendering uses a synthesised voice; it shows a structure, not a voice.

- The areas, the feelings and the sorting of practices are rough categories.
- The scaffolding claim is broad, and can only be tested area by area.
- The established findings cited here show that the materials exist; none of them tests the construction.
- The text was written by AI (below). Every reference was taken from its own record, but errors of emphasis or reading are possible, and corrections are welcome.

---

## 12. Safety

Meditation and related practices can harm. Across 83 studies, adverse events were found in about 8% of those who meditated (about 4% in experimental studies and 33% in observational ones), anxiety and depression most often, sometimes in people with no previous mental-health problems (Farias et al. 2020). In the Varieties of Contemplative Experience study, which recruited practitioners who could describe challenging or difficult experiences, symptoms lasted a median of one to three years, from a few days to more than ten (Lindahl et al. 2017). Attention to the body is not always benign: there are signs that past some point its effects turn negative (Britton 2019).

Breathwork carries its own risks. In one trained method, studied in hospital under monitoring, cycles of fast breathing and breath-holding brought the blood's carbon dioxide down and then its oxygen, whose saturation fell for some seconds to around 50% (Kox et al. 2014). Done before swimming underwater, fast breathing has made healthy swimmers lose consciousness and drown (Boyd et al. 2015). Immersion in cold water can lead to drowning, cardiac arrest and hypothermia (Tipton et al. 2017). Straining against a held breath adds to the rise in blood pressure that heavy effort brings: in five experienced bodybuilders lifting to failure, blood pressure peaked at a mean of 320/250 mmHg (MacDougall et al. 1985).

Nothing in this paper is a practice, a protocol or a recommendation, and nothing in it is medical advice. The claims about bodily function and healing are hypotheses; anyone with a problem in these areas should see a physician. Anyone in difficulty with meditation can find support from Cheetah House. Anyone thinking about suicide should reach out now: in the US, call or text 988; elsewhere, find a line at findahelpline.com. Any study with people belongs under ethics review.

---

## 13. Conclusion

OM is among the oldest and simplest-sounding of chants. The hypothesis here is that its simplicity is a surface. Beneath one long tone lies a fast sequence of notes made in many ways, arranged in permuted sections, and carried by a breath of thousands of tiny turns. Beneath meditation lies the same breath, inverted.

If this is right, the deepest meditation would belong to those who can build OM most perfectly, and every practice would be a partial form of one construction. It can be shown wrong cheaply. If the breath carries no fine structure beyond the heartbeat's, if a skilled OM's pitch moves as one line, or if a plain hum does what a well-built OM does, it should be set aside.

**Box 1. Open questions**

- How fast and how deep are the tiny breaths, and how are they spread across one large breath?
- Do the simpler sections come first or last, and does it differ between the types?
- What sets the drone apart from the curve?
- How does the athletic breath divide an area into functions?
- Do practices from different traditions share the fine structure of their area?

## Authorship and AI use

This paper was written by AI under the author's direction. Eugene Jhong conceived the hypothesis and the construction of OM, developed in the OM Project, and supplied the practitioner observations behind it. The text was drafted by Claude Opus 5.5 (Anthropic) from conversations with the author, the project's score and its recordings. A draft was then refereed by a second model, Claude Fable 5.1 (Anthropic), which checked each claim against its source; its corrections were checked again against the sources before they were made. Every reference was taken from its own record (PubMed's for papers indexed there, and Crossref's and Open Library's for the books), not from a model's memory. The author is responsible for the content.

## Data and code

The OM Project's score and player are open at ejhong.github.io/om, with code at github.com/ejhong/om. Figure 1b was computed from the author's recordings with the pYIN pitch tracker in librosa.

## Competing interests and funding

None.

## Versions

Draft 1 (6 November 2026): the first version.

## References


Adler TE, Coovadia Y, Cirone D et al. (2019). Device-guided slow breathing reduces blood pressure and sympathetic activity in young normotensive individuals of both sexes. J Appl Physiol 127:1042–1049. doi:10.1152/japplphysiol.00442.2019

Arsenault M, Ladouceur A, Lehmann A et al. (2013). Pain modulation induced by respiration: phase and frequency effects. Neuroscience 252:501–511. doi:10.1016/j.neuroscience.2013.07.048

Bernardi L, Sleight P, Bandinelli G et al. (2001). Effect of rosary prayer and yoga mantras on autonomic cardiovascular rhythms: comparative study. BMJ 323:1446–1449. doi:10.1136/bmj.323.7327.1446

Boyd C, Levy A, McProud T et al. (2015). Fatal and nonfatal drowning outcomes related to dangerous underwater breath-holding behaviors: New York State, 1988–2011. MMWR Morb Mortal Wkly Rep 64:518–521. PubMed 25996093

Bramble DM, Carrier DR (1983). Running and breathing in mammals. Science 219:251–256. doi:10.1126/science.6849136

Britton WB (2019). Can mindfulness be too much of a good thing? The value of a middle way. Curr Opin Psychol 28:159–165. doi:10.1016/j.copsyc.2018.12.011

Busch V, Magerl W, Kern U et al. (2012). The effect of deep and slow breathing on pain perception, autonomic activity, and mood processing: an experimental study. Pain Med 13:215–228. doi:10.1111/j.1526-4637.2011.01243.x

Cheetah House. Support for meditators and their families. cheetahhouse.org

Farias M, Maraldi E, Wallenkampf KC et al. (2020). Adverse events in meditation practices and meditation-based therapies: a systematic review. Acta Psychiatr Scand 142:374–393. doi:10.1111/acps.13225

Hodges PW, Gandevia SC (2000). Changes in intra-abdominal pressure during postural and respiratory activation of the human diaphragm. J Appl Physiol 89:967–976. doi:10.1152/jappl.2000.89.3.967

Jafari H, Courtois I, Van den Bergh O et al. (2017). Pain and respiration: a systematic review. Pain 158:995–1006. doi:10.1097/j.pain.0000000000000865

Jhong E (2026). The perforator hypothesis: knots as small arteries held shut where they pierce the fascia. Zenodo preprint. doi:10.5281/zenodo.23901648

Kalyani BG, Venkatasubramanian G, Arasappa R et al. (2011). Neurohemodynamic correlates of 'OM' chanting: a pilot functional magnetic resonance imaging study. Int J Yoga 4:3–6. doi:10.4103/0973-6131.78171

Kox M, van Eijk LT, Zwaag J et al. (2014). Voluntary activation of the sympathetic nervous system and attenuation of the innate immune response in humans. Proc Natl Acad Sci U S A 111:7379–7384. doi:10.1073/pnas.1322174111

Kozhevnikov M, Elliott J, Shephard J et al. (2013). Neurocognitive and somatic components of temperature increases during g-tummo meditation: legend and reality. PLoS One 8:e58244. doi:10.1371/journal.pone.0058244

Lindahl JR, Fisher NE, Cooper DJ et al. (2017). The varieties of contemplative experience: a mixed-methods study of meditation-related challenges in Western Buddhists. PLoS One 12:e0176239. doi:10.1371/journal.pone.0176239

MacDougall JD, Tuxen D, Sale DG et al. (1985). Arterial blood pressure response to heavy resistance exercise. J Appl Physiol 58:785–790. doi:10.1152/jappl.1985.58.3.785

Mallinson J, Singleton M (2017). Roots of Yoga. Penguin Classics. ISBN 9780241253045

Mürbe D, Zahnert T, Kuhlisch E et al. (2007). Effects of professional singing education on vocal vibrato: a longitudinal study. J Voice 21:683–688. doi:10.1016/j.jvoice.2006.06.002

O'Connell DG, Hinman MR, Hearne KF et al. (2014). The effects of "grunting" on serve and forehand velocities in collegiate tennis players. J Strength Cond Res 28:3469–3475. doi:10.1519/JSC.0000000000000604

Olivelle P (1998). The Early Upanisads. Oxford University Press. Mandukya Upanisad, pp. 473–477. doi:10.1093/oso/9780195124354.003.0013

Oneda B, Ortega KC, Gusmão JL et al. (2010). Sympathetic nerve activity is decreased during device-guided slow breathing. Hypertens Res 33:708–712. doi:10.1038/hr.2010.74

Perl O, Ravia A, Rubinson M et al. (2019). Human non-olfactory cognition phase-locked with inhalation. Nat Hum Behav 3:501–512. doi:10.1038/s41562-019-0556-z

Roubeau B, Henrich N, Castellengo M (2009). Laryngeal vibratory mechanisms: the notion of vocal register revisited. J Voice 23:425–438. doi:10.1016/j.jvoice.2007.10.014

Sundberg J (2001). Level and center frequency of the singer's formant. J Voice 15:176–186. doi:10.1016/S0892-1997(01)00019-4

Thomasson M, Sundberg J (1999). Consistency of phonatory breathing patterns in professional operatic singers. J Voice 13:529–541. doi:10.1016/s0892-1997(99)80008-3

Tipton MJ, Collier N, Massey H et al. (2017). Cold water immersion: kill or cure? Exp Physiol 102:1335–1355. doi:10.1113/EP086283

Watson PJ, Hixon TJ (1985). Respiratory kinematics in classical (opera) singers. J Speech Hear Res 28:104–122. doi:10.1044/jshr.2801.104

Zautra AJ, Fasman R, Davis MC et al. (2010). The effects of slow breathing on affective responses to pain stimuli: an experimental study. Pain 149:12–18. doi:10.1016/j.pain.2009.10.001

Zelano C, Jiang H, Zhou G et al. (2016). Nasal respiration entrains human limbic oscillations and modulates cognitive function. J Neurosci 36:12448–12467. doi:10.1523/JNEUROSCI.2586-16.2016